Elastic London Meetup

Mon Sept 15, 2025 | 6:30PM

Carly Richmond

elastic | The Search AI Company

What’s New?

Elastic 8.19/9.1 🎉

- GA of Cross Cluster Search with ES|QL, Enterprise
- LOOKUP JOINs with ES|QL
- LogsDB & TSDS indexing mode optimisations
- BBQ enabled by default
- ACORN filtered vector search

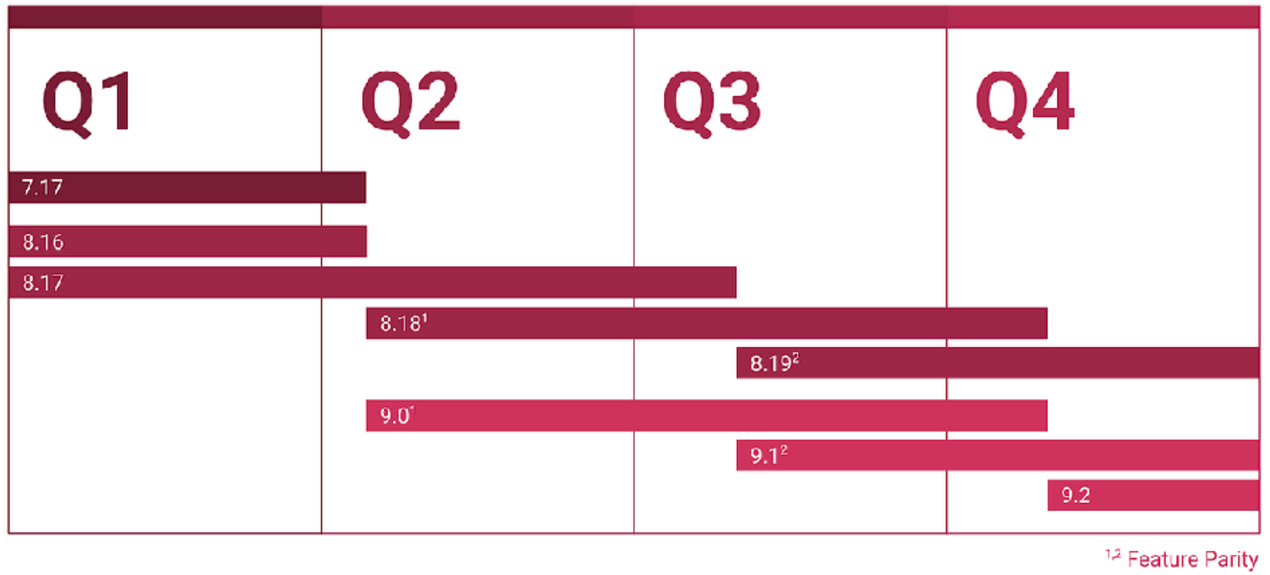
```
FROM kibana_sample_data_logs | WHERE response.keyword != "200"
| LOOKUP JOIN teams_lkp ON host
| STATS num = COUNT(*) by host, response.keyword, team
| SORT num DESC
```

https://discuss.elastic.co/t/what-s-new-in-elastic-8-19-9-1

Elastic Stack Releases & Maintenance

Elastic 8.19/9.1 🎉

- Approximately quarterly releases, with 2 minor versions maintained in parallel, to give you more time for upgrades and provide an overlap in maintained versions.
- 8.19 will be maintained for quite a bit longer, but from 9.2 on, all new features will only make it to the 9.x series.
- If you are still on 7.x or earlier, now is a great time to plan moving to a maintained version.



https://discuss.elastic.co/t/what-s-new-in-elastic-8-19-9-1

elastic

Doing something cool with elastic?

Speak at our meetups!

https://sessionize.com/elastic-meetups/

Doing something cool with elastic?

ELASTIC{ON} is coming to London!

26 February 2026
Convene Sancroft, St. Paul's

Register:
https://www.elastic.co/events/elasticon/london

https://sessionize.com/elasticon/

Carly Richmond

Developer Advocate

Lead, elastic

Alam Ahmed

Cloud & DevOps Engineer

elastic

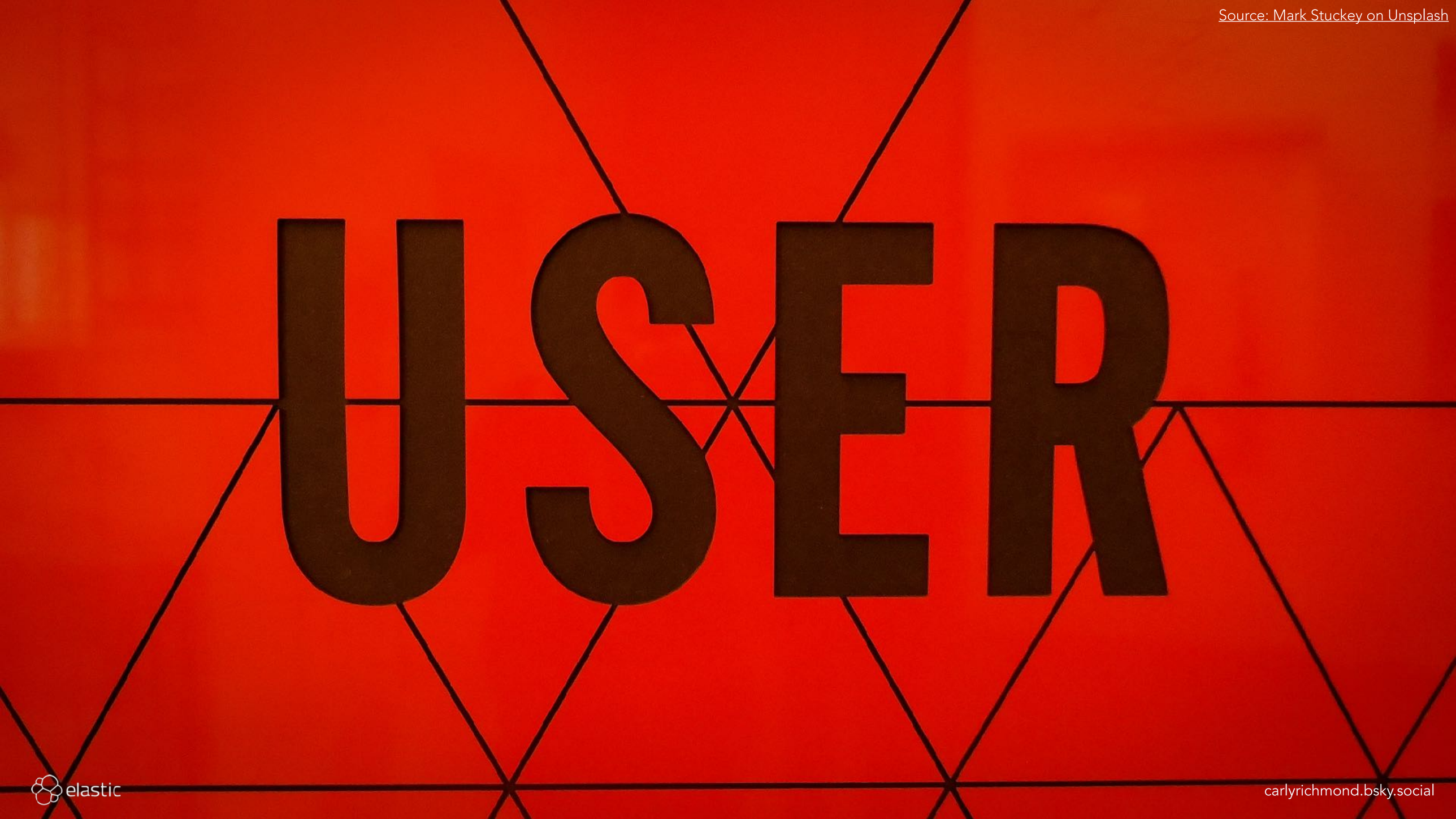

Source: Patrick Hendry on Unsplash
WEB UI INSTRUMENTATION WITH
OpenTelemetry & elastic
CARLY RICHMOND
elastic
carlyrichmond.bsky.social

ABOUT ME

- Developer Advocate Lead @ elastic
- Frontend Engineer, Speaker & Blogger

SCAN ME

carlyrichmond.bsky.social



Source: Mark Stuckey on Unsplash

FIRST PORT OF CALL

Source: Dominik Lückmann on Unsplash

BACK OF THE QUEUE

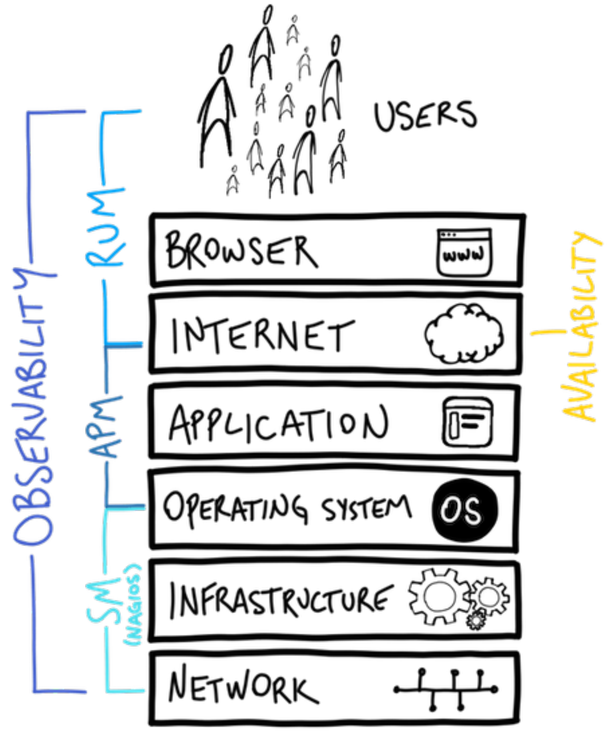
Source: Rach Teo on Unsplash

REAL USER MONITORING

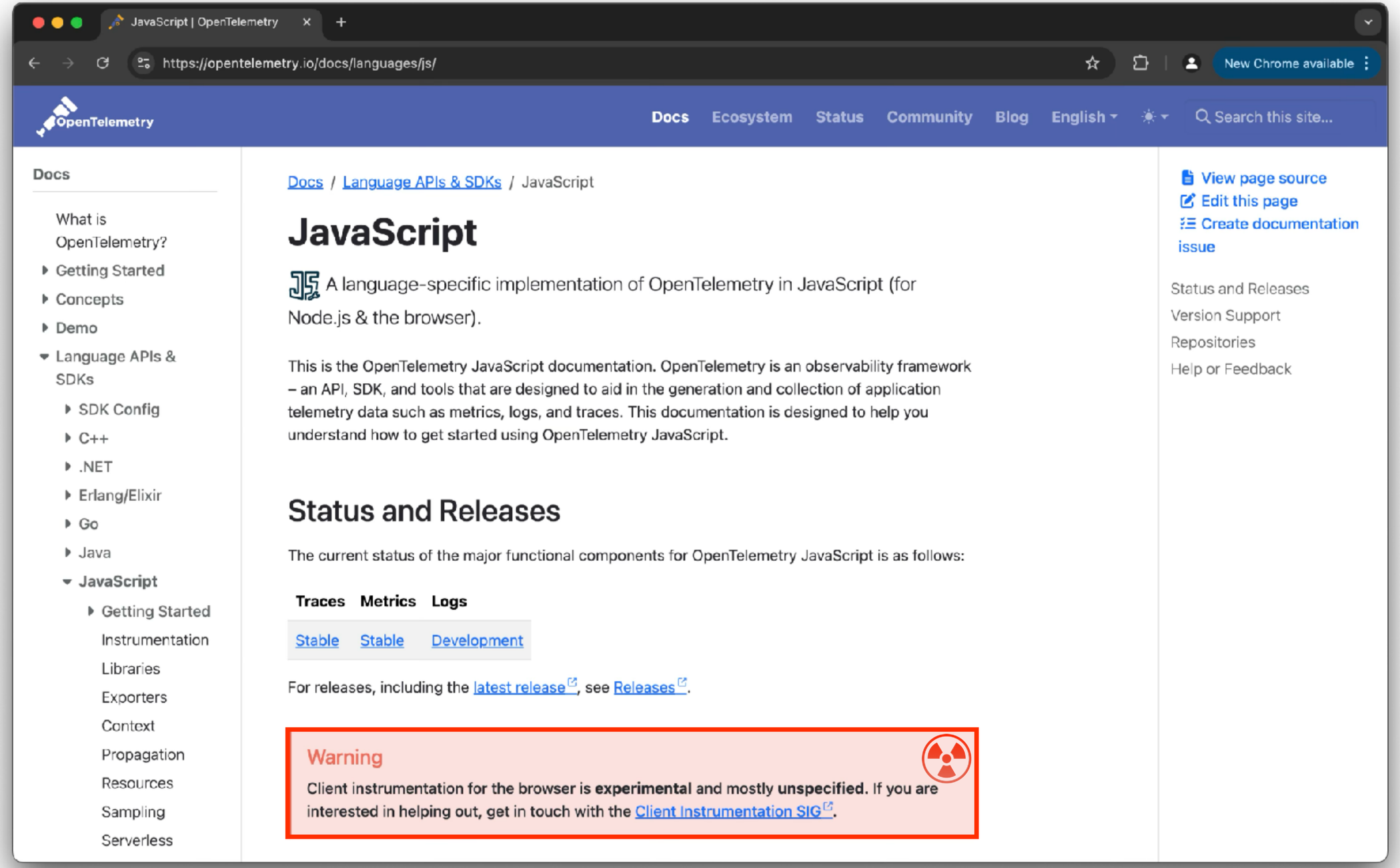
STILL ON THE RUM?

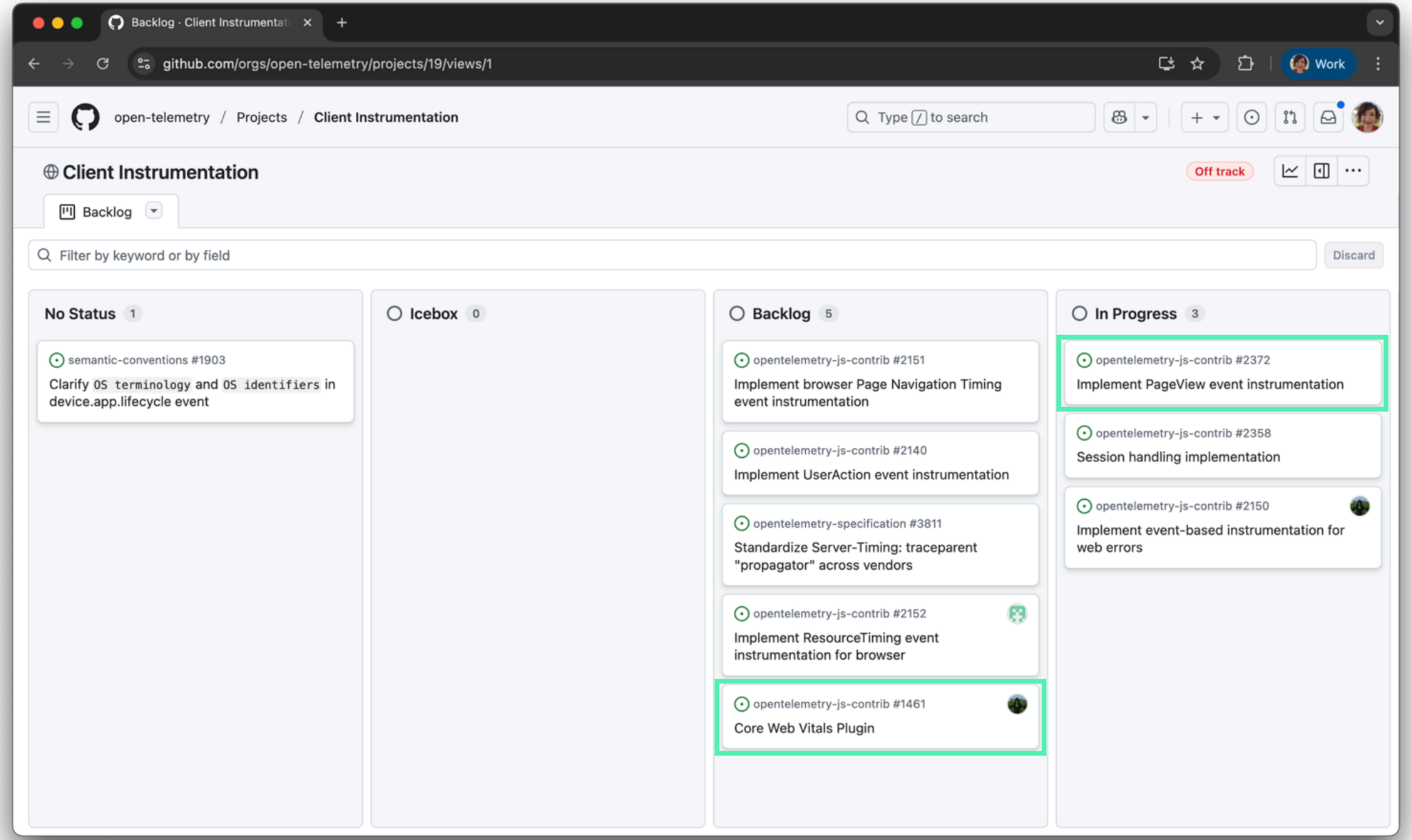
Source: Meta Bakrate on Unsplash

FRONT TO BACK (OPEN) TELEMETRY



https://www.stack.io/blog/a-history-of-monitoring-and-opentelemetry


JavaScript | OpenTelemetry
https://opentelemetry.io/docs/languages/js/
New Chrome available
OpenTelemetry
Docs
Ecosystem
Status
Community
Blog
English
Search this site...
Docs
What is OpenTelemetry?
Getting Started
Concepts
Demo
Language APIs & SDKs
SDK Config
C++
.NET
Erlang/Elixir
Go
Java
JavaScript
Getting Started
Instrumentation
Libraries
Exporters
Context
Propagation
Resources
Sampling
Serverless
Docs / Language APIs & SDKs / JavaScript
JavaScript
A language-specific implementation of OpenTelemetry in JavaScript (for Node.js & the browser).
This is the OpenTelemetry JavaScript documentation. OpenTelemetry is an observability framework – an API, SDK, and tools that are designed to aid in the generation and collection of application telemetry data such as metrics, logs, and traces. This documentation is designed to help you understand how to get started using OpenTelemetry JavaScript.
Status and Releases
The current status of the major functional components for OpenTelemetry JavaScript is as follows:
Traces
Metrics
Logs
Stable
Stable
Development
For releases, including the latest release, see Releases.
Warning
Client instrumentation for the browser is experimental and mostly unspecified. If you are interested in helping out, get in touch with the Client Instrumentation SIG.
View page source
Edit this page
Create documentation issue
Status and Releases
Version Support
Repositories
Help or Feedback


Backlog · Client Instrumentati
github.com/orgs/open-telemetry/projects/19/views/1
Work
open-telemetry / Projects / Client Instrumentation
Type / to search
Client Instrumentation
Off track
Backlog
Filter by keyword or by field
Discard
No Status 1
semantic-conventions #1903
Clarify OS terminology and OS identifiers in device.app.lifecycle event
Icebox 0
Backlog 5
opentelemetry-js-contrib #2151
Implement browser Page Navigation Timing event instrumentation
opentelemetry-js-contrib #2140
Implement UserAction event instrumentation
opentelemetry-specification #3811
Standardize Server-Timing: traceparent "propagator" across vendors
opentelemetry-js-contrib #2152
Implement ResourceTiming event instrumentation for browser
opentelemetry-js-contrib #1461
Core Web Vitals Plugin
In Progress 3
opentelemetry-js-contrib #2372
Implement PageView event instrumentation
opentelemetry-js-contrib #2358
Session handling implementation
opentelemetry-js-contrib #2150
Implement event-based instrumentation for web errors

localhost:4173

localhost:4173

Finish update

OTel Records

Records Events News

Featured

Loveless
My Bloody Valentine
Available formats

Licensed to Ill
Beastie Boys
Available formats

Hejira
Joni Mitchell
Available formats

Goo
Sonic Youth
Available formats

The Rise and Fall..
David Bowie
Available formats

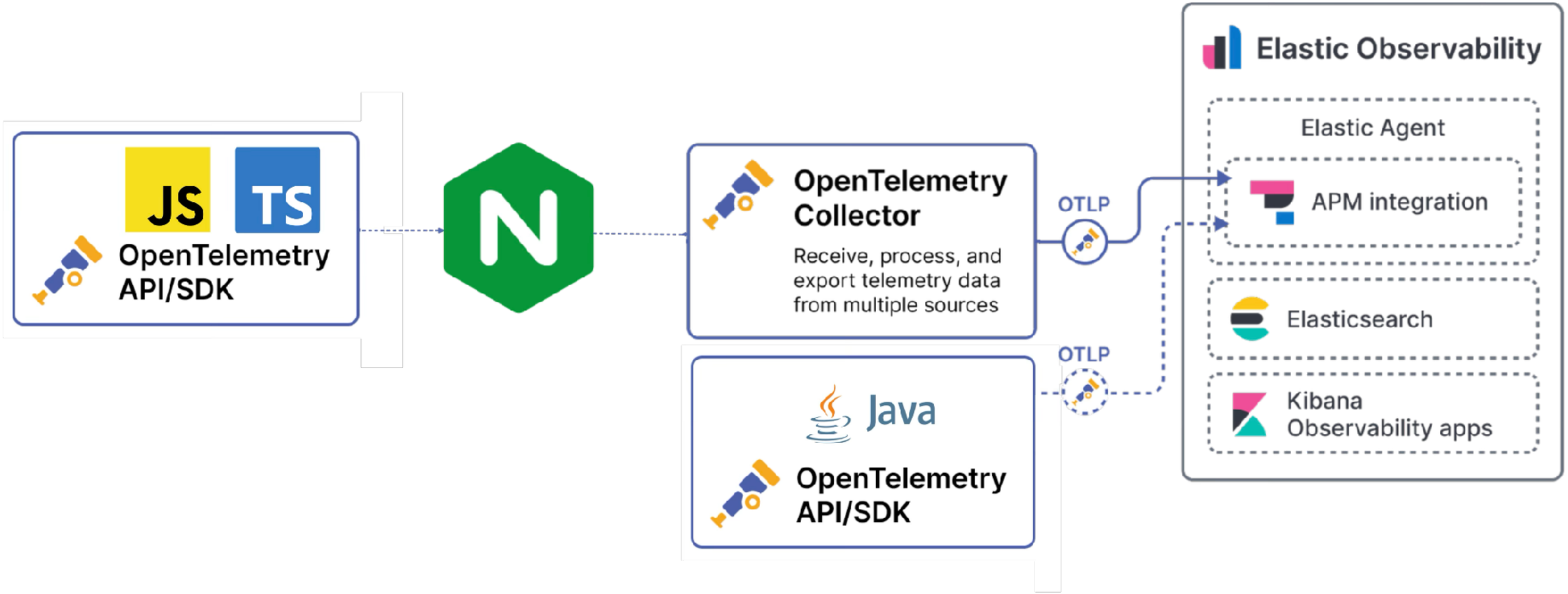
Made by Carly Richmond with 💜 and excessive amounts of 🍵

SCAN ME

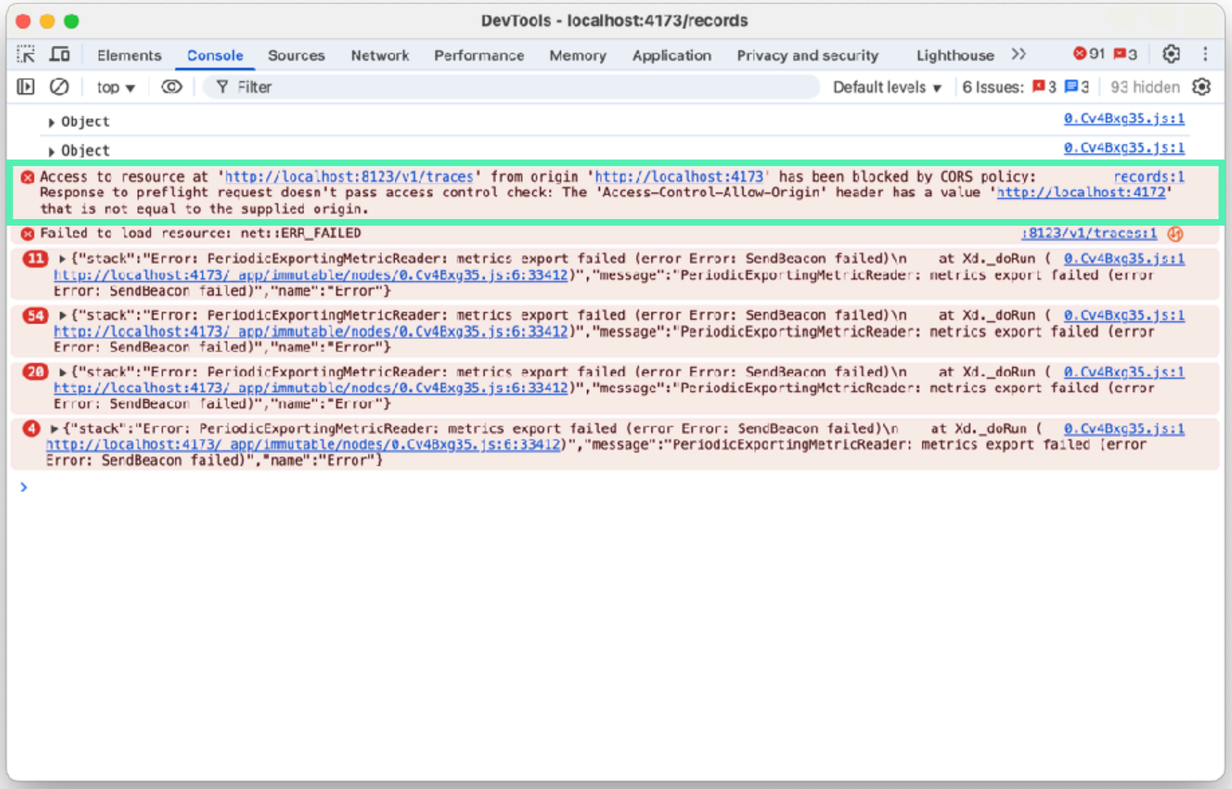
elastic

carlyrichmond.bsky.social

OTEL WEB ARCHITECTURE



CUE CORS HELL!



NGINX PROXY

nginx.conf

```
events {}

http {

  server {

    listen 8123;

    location /v1/traces {
      proxy_pass http://host.docker.internal:4318;
      # Apply CORS headers to ALL responses, including POST
      add_header 'Access-Control-Allow-Origin' 'http://localhost:4173' always;
      add_header 'Access-Control-Allow-Methods' 'POST, OPTIONS' always;
      add_header 'Access-Control-Allow-Headers' 'Content-Type' always;
      add_header 'Access-Control-Allow-Credentials' 'true' always;

      # Preflight requests receive a 204 No Content response
      if ($request_method = OPTIONS) {
        return 204;
      }
    }
  }
}
```

frontend.tracer.ts

```
/* OpenTelemetry Frontend packages */

// See https://github.com/carlyrichmond/otel-record-store for import list

// Define resource metadata for the service, used by exporters (Elastic requires service.name)
const SERVICE_NAME = 'records-ui-web';

const detectedResources = detectResources({ detectors: [browserDetector] });
let resource = resourceFromAttributes({
    [ATTR_SERVICE_NAME]: SERVICE_NAME,
    'service.version': 1,
    'deployment.environment': 'dev'
});
resource = resource.merge(detectedResources);

// Configure logging to send to the collector via nginx
const logExporter = new OTLPLogExporter({
    url: LOGS_URL // nginx proxy
});

const loggerProvider = new LoggerProvider({
    resource: resource,
    processors: [new BatchLogRecordProcessor(logExporter)]
});

logs.setGlobalLoggerProvider(loggerProvider);

```

Source: Ruben. on Unsplash

Logs

Metrics

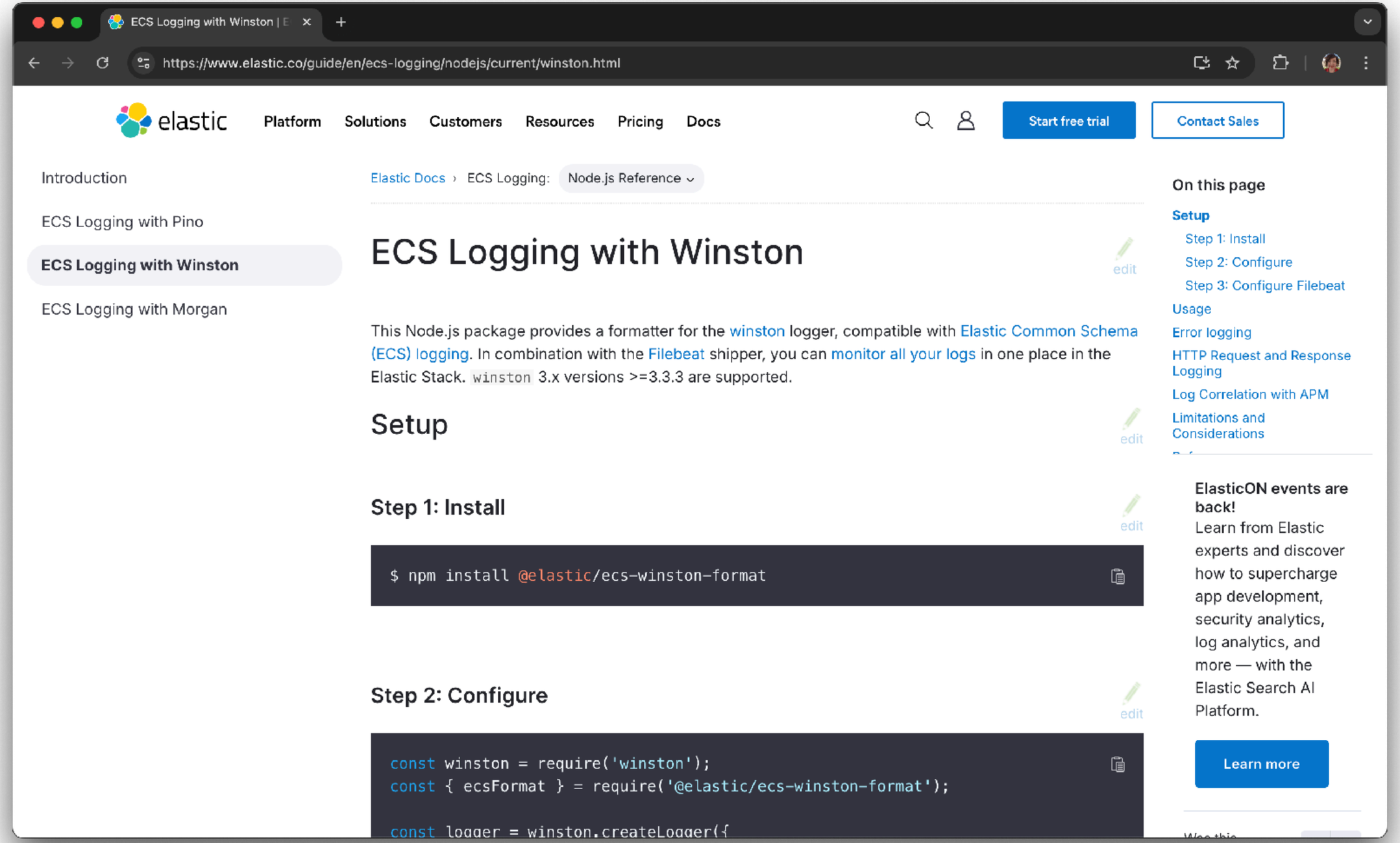
Traces

Source: Ruben. on Unsplash

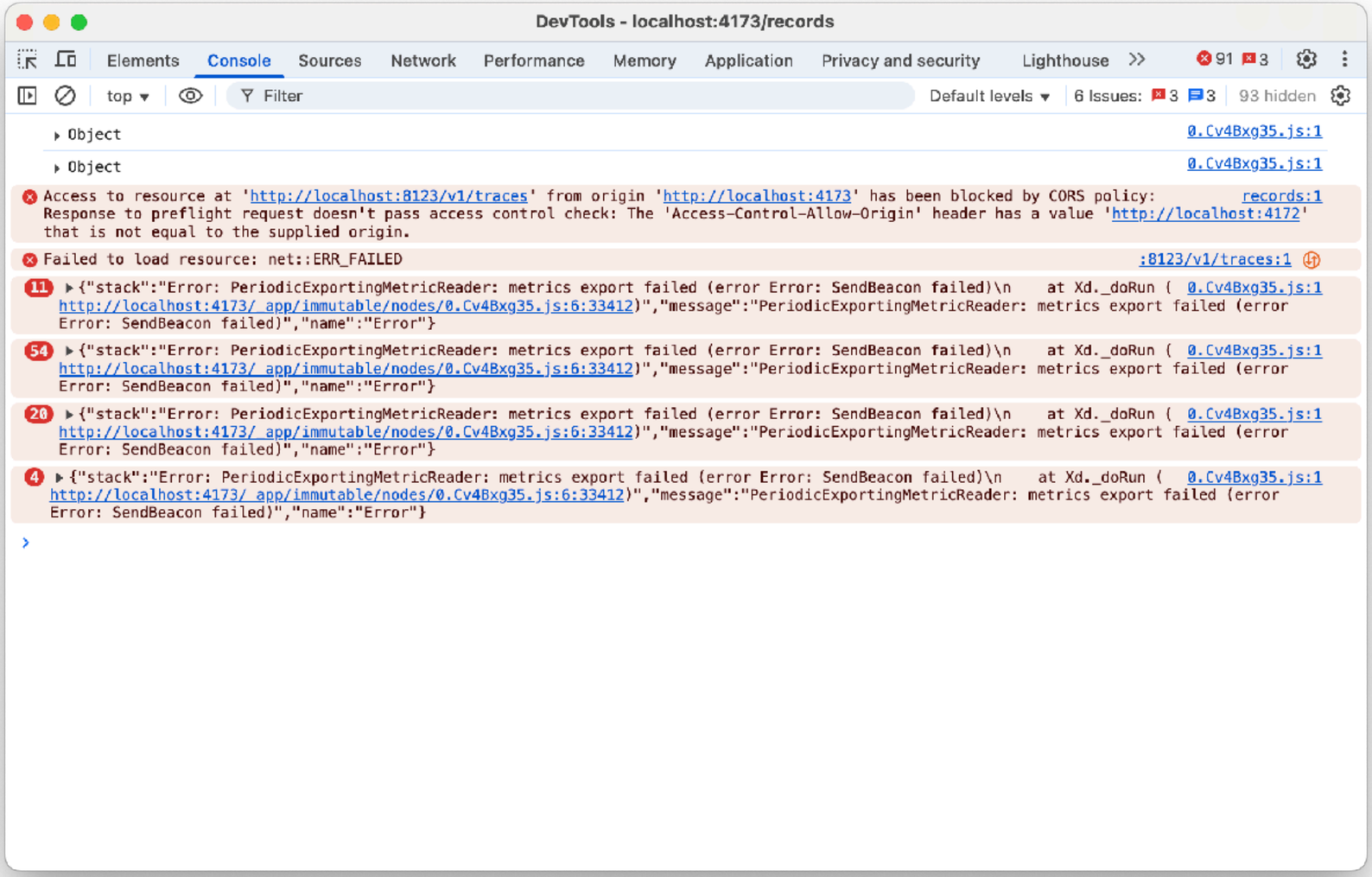
Logs

Metrics

Traces


ECS Logging with Winston | E
https://www.elastic.co/guide/en/ecs-logging/nodejs/current/winston.html
elastic
Platform
Solutions
Customers
Resources
Pricing
Docs
Start free trial
Contact Sales
Introduction
ECS Logging with Pino
ECS Logging with Winston
ECS Logging with Morgan
Elastic Docs › ECS Logging: Node.js Reference
ECS Logging with Winston
edit
This Node.js package provides a formatter for the winston logger, compatible with Elastic Common Schema (ECS) logging. In combination with the Filebeat shipper, you can monitor all your logs in one place in the Elastic Stack. winston 3.x versions >=3.3.3 are supported.
Setup
edit
Step 1: Install
edit
$ npm install @elastic/ecs-winston-format
Step 2: Configure
edit
const winston = require('winston');
const { ecsFormat } = require('@elastic/ecs-winston-format');
const logger = winston.createLogger({
On this page
Setup
Step 1: Install
Step 2: Configure
Step 3: Configure Filebeat
Usage
Error logging
HTTP Request and Response Logging
Log Correlation with APM
Limitations and Considerations
ElasticON events are back!
Learn from Elastic experts and discover how to supercharge app development, security analytics, log analytics, and more — with the Elastic Search AI Platform.
Learn more

DEVTOOLS CONSOLE ISN'T VISIBLE!



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logs.setGlobalLoggerProvider(loggerProvider);

```

```
11      'service.version': 1,
12      'deployment.environment': 'dev'
13  });
14  resource = resource.merge(detectedResources);
15
16  // Configure logging to send to the collector via nginx
17  const logExporter = new OTLPLogExporter({
18      url: LOGS_URL // nginx proxy
19  });
20
21  const loggerProvider = new LoggerProvider({
22      resource: resource,
23      processors: [new BatchLogRecordProcessor(logExporter)]
24  });
25
26  logs.setGlobalLoggerProvider(loggerProvider);
27
28  const logger = logs.getLogger('default', '1.0.0');
29  logger.emit({
30      severityNumber: SeverityNumber.INFO,
31      severityText: 'INFO',
32      body: 'Logger initialized'
33  });
34
35  // Configure the OTLP exporter to talk to the collector via nginx
36  const exporter = new OTLPTraceExporter({
37      url: TRACE_URL // nginx proxy
38  });
39
```

Data view: All logs

service.name : "records-ui-web"

Last 15 minutes | Refresh

Search field names

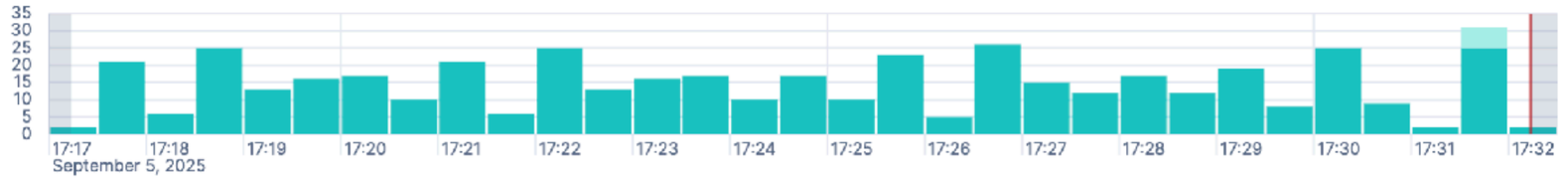
Selected fields 3
- body.text
- severity_text
- severity_number

Popular fields 3
- body.text
- severity_number
- severity_text

Available fields 133
- @timestamp
- agent.name
- agent.version
- attributes.error.exception.handled
- attributes.error.grouping_key
- attributes.error.grouping_name
- attributes.error.id
- attributes.event.name
- attributes.exception.message
- attributes.exception.type
- attributes.exit_code
- attributes.http.conn.done.addr

Add a field

Auto interval | Breakdown by log.level



Sep 5, 2025 @ 17:17:13.637 - Sep 5, 2025 @ 17:32:13.637 (interval: Auto - 30 seconds)

Documents (451) | Patterns | Field statistics

Columns 4 | Sort fields 1

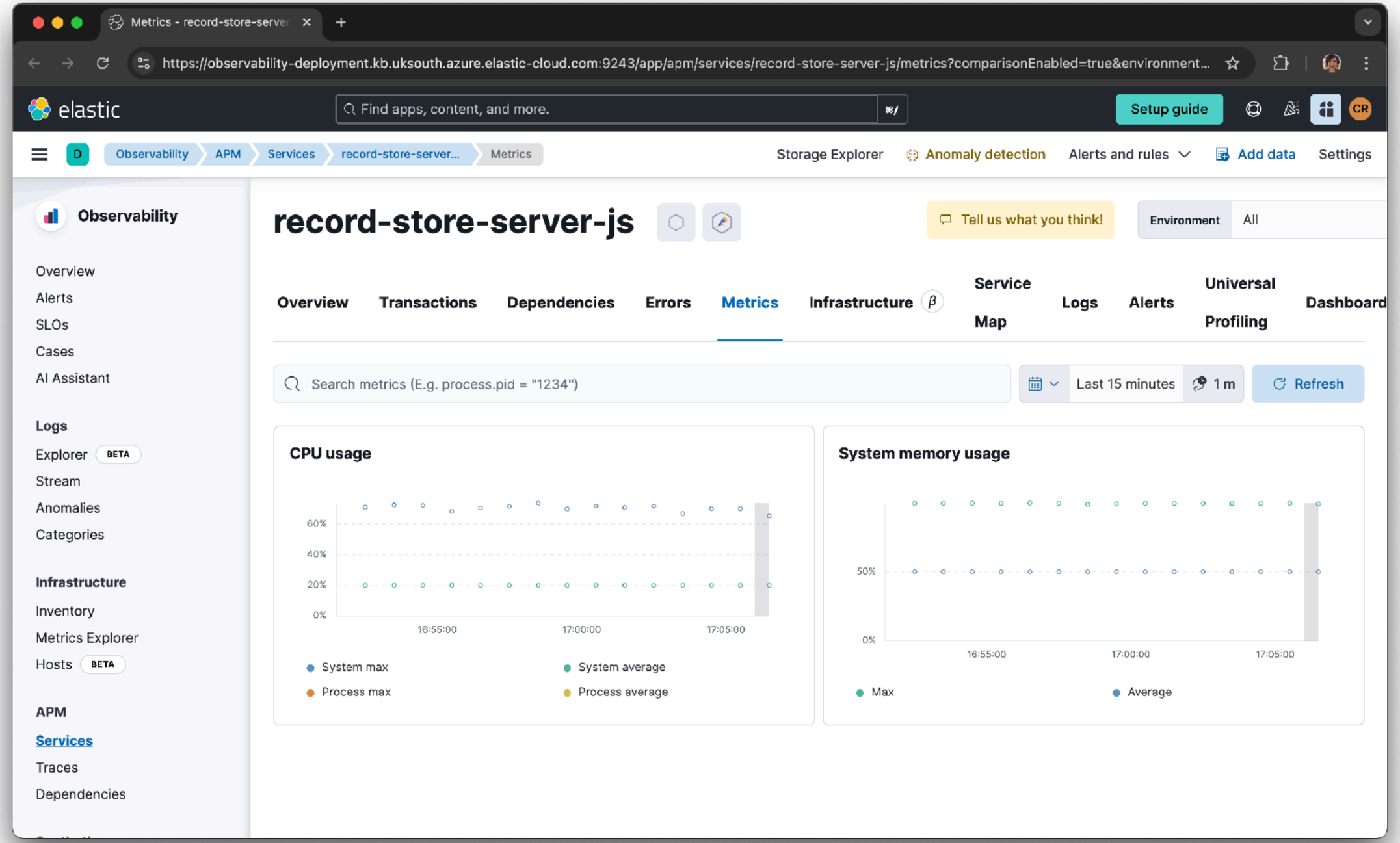
Actions	@timestamp	body.text	severity_text	severity_number
	Sep 5, 2025 @ 17:32:03.305	Client Telemetry Initialised	INFO	9
	Sep 5, 2025 @ 17:32:03.296	Logger initialized	INFO	9
	Sep 5, 2025 @ 17:31:58.489	Client Telemetry Initialised	INFO	9
	Sep 5, 2025 @ 17:31:58.480	Logger initialized	INFO	9
	Sep 5, 2025 @ 17:31:57.736	Client Telemetry Initialised	INFO	9
	Sep 5, 2025 @ 17:31:57.727	Logger initialized	INFO	9
	Sep 5, 2025 @ 17:31:53.188	Client Telemetry Initialised	INFO	9

Source: Ruben. on Unsplash

Logs

Metrics

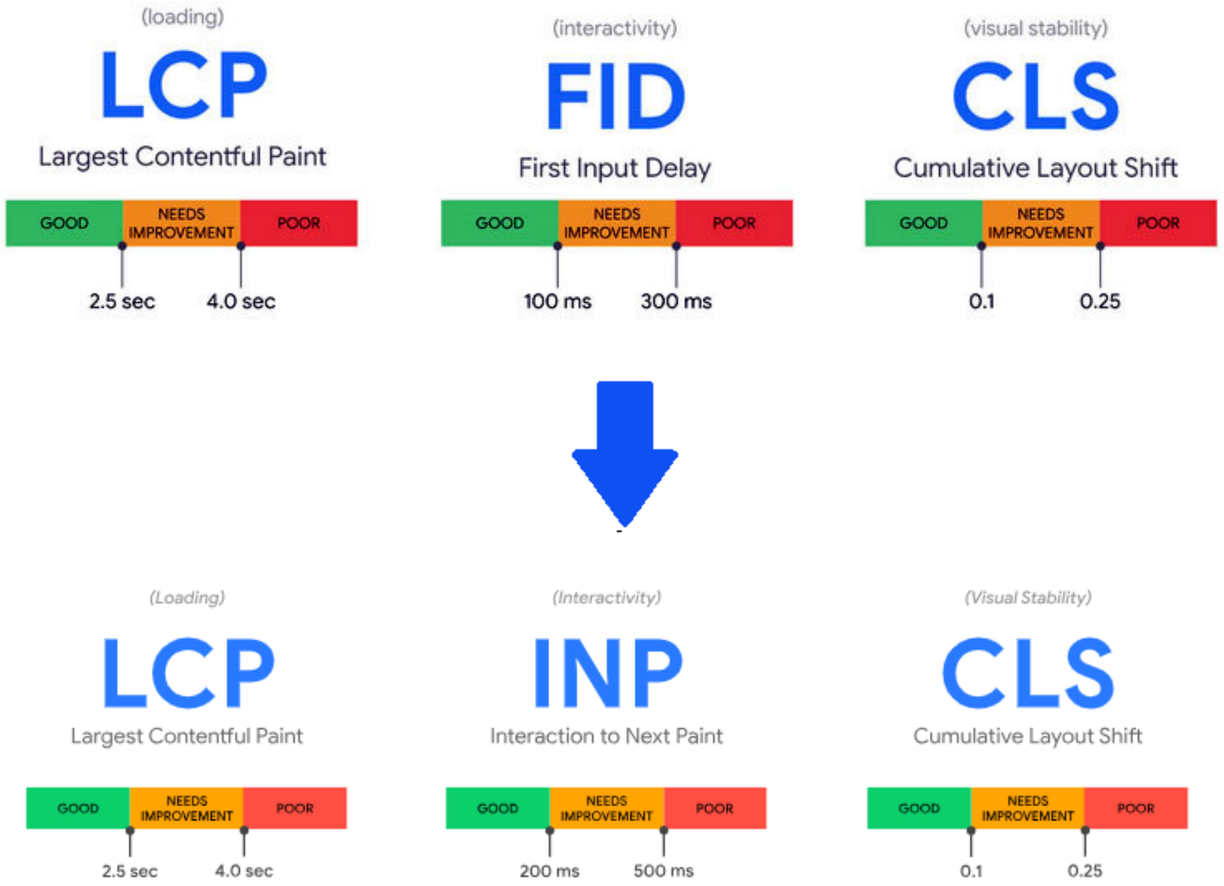
Traces


Metrics - record-store-server
https://observability-deployment.kb.uksouth.azure.elastic-cloud.com:9243/app/apm/services/record-store-server-js/metrics?comparisonEnabled=true&environment...
elastic
Find apps, content, and more.
Setup guide
CR
D
Observability
APM
Services
record-store-server...
Metrics
Storage Explorer
Anomaly detection
Alerts and rules
Add data
Settings
Observability
Overview
Alerts
SLOs
Cases
AI Assistant
Logs
Explorer BETA
Stream
Anomalies
Categories
Infrastructure
Inventory
Metrics Explorer
Hosts BETA
APM
Services
Traces
Dependencies
record-store-server-js
Tell us what you think!
Environment
All
Overview
Transactions
Dependencies
Errors
Metrics
Infrastructure
Service Map
Logs
Alerts
Universal Profiling
Dashboard
Search metrics (E.g. process.pid = "1234")
Last 15 minutes
1 m
Refresh
CPU usage
60%
40%
20%
0%
16:55:00
17:00:00
17:05:00
System max
System average
Process max
Process average
System memory usage
50%
0%
16:55:00
17:00:00
17:05:00
Max
Average

Source: Unsplash

https://web.dev/explore/learn-core-web-vitals

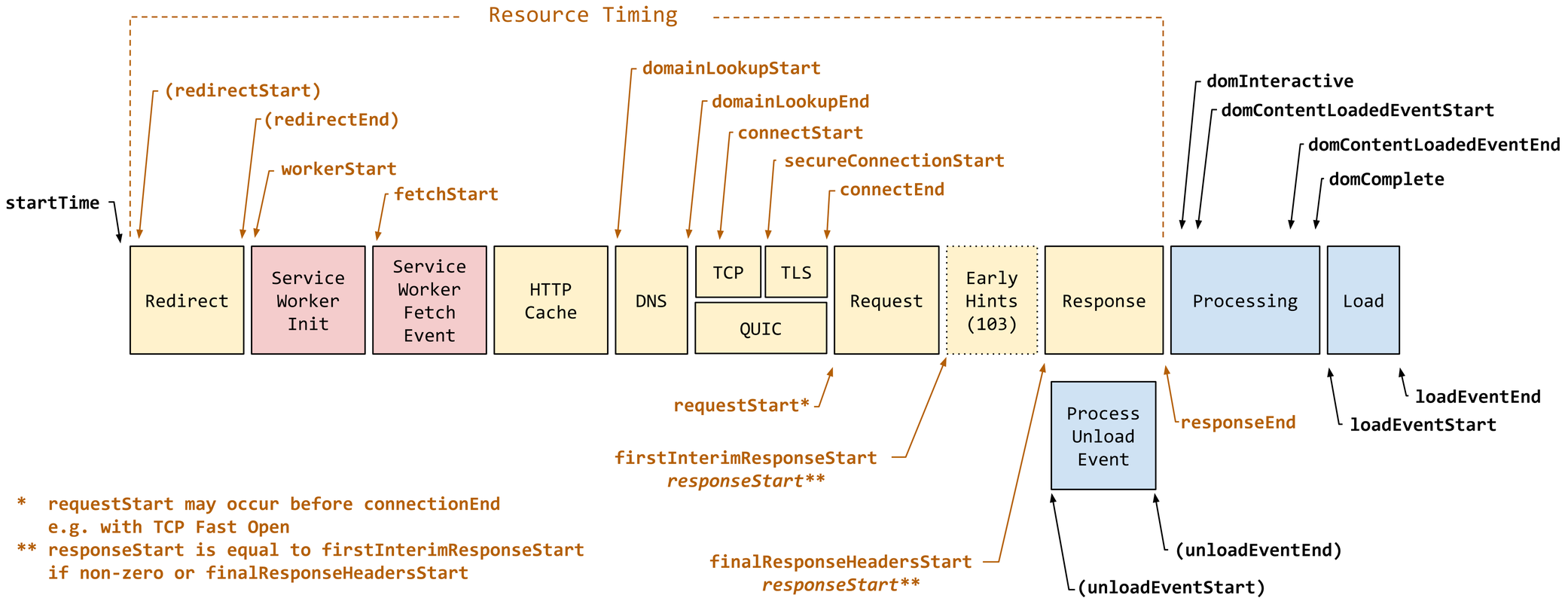
GOOGLE CORE WEB VITALS



https://web.dev/explore/learn-core-web-vitals

OTHER VITALS

PAGE LOAD

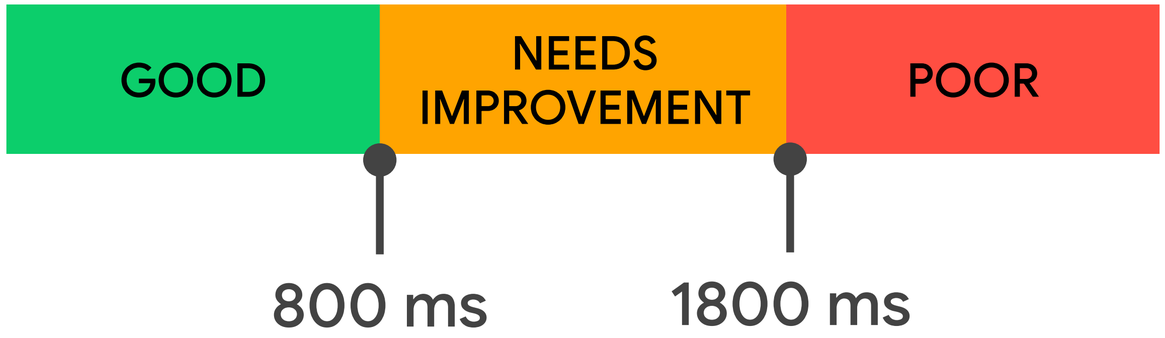


https://web.dev/articles/ttfb

TTFB

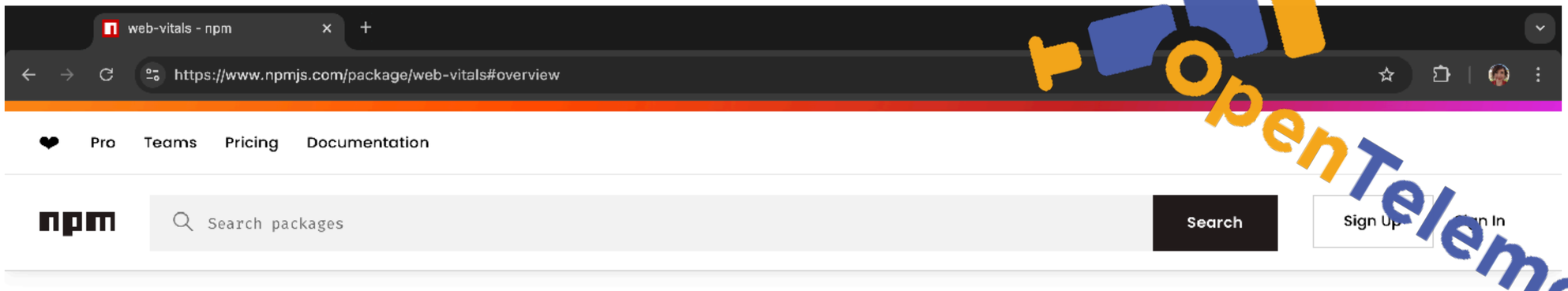
TIME TO FIRST BYTE

The time between the request for a resource being initiated and when the first byte of a response begins to arrive.



https://web.dev/articles/ttfb

Source: Ruben. on Unsplash



OpenTelemetry

web-vitals TS

4.2.3 • Public • Published 2 months ago

Readme | Code Beta | 0 Dependencies | 18,887 Dependents | 76 Versions

web-vitals

- Overview
- Install and load the library
  - From npm
  - From a CDN
- Usage
  - Basic usage
  - Report the value on every change
  - Report only the delta of changes
  - Send the results to an analytics endpoint
  - Send the results to Google Analytics
  - Send the results to Google Tag Manager
  - Send attribution data
  - Batch multiple reports together

Install

```
> npm i web-vitals
```

Repository

github.com/GoogleChrome/web-vitals

Homepage

github.com/GoogleChrome/web-vitals#...

Weekly Downloads

4,090,498

Version	License
4.2.3	Apache-2.0

Unpacked Size	Total Files

web-vitals.instrumentation.ts

```ts
/* Web Vitals Frontend package*/

// See https://github.com/carlyrichmond/otel-record-store for import list

type CWVMetric = LCPMetric | CLSMetric | INPMetric | TTFBMetric | FCPMetric;

export class WebVitalsInstrumentation extends InstrumentationBase {

    private cwvMeter: Meter;

    /* Core Web Vitals Measures */
    private lcp: ObservableGauge;
    private cls: ObservableGauge;
    private inp: ObservableGauge;
    private ttfb: ObservableGauge;
    private fcp: ObservableGauge;

    constructor(config: InstrumentationConfig, resource: Resource) {
        super('WebVitalsInstrumentation', '1.0', config);

        const myServiceMeterProvider = this.generateMeterForResource(resource);
        metrics.setGlobalMeterProvider(myServiceMeterProvider);

        this.cwvMeter = metrics.getMeter('core-web-vitals', '1.0.0');

        // Initialising CWV metric instruments
        this.lcp = this.cwvMeter.createObservableGauge('lcp', { unit: 'ms', description: 'Largest Contentful Paint' });
        this.cls = this.cwvMeter.createObservableGauge('cls', { description: 'Cumulative Layout Shift' });
        this.inp = this.cwvMeter.createObservableGauge('inp', { unit: 'ms', description: 'Interaction to Next Paint' });
        this.ttfb = this.cwvMeter.createObservableGauge('ttfb', { unit: 'ms', description: 'Time to First Byte' });
```

```
18    constructor(config: InstrumentationConfig, resource: Resource) {
19        super('WebVitalsInstrumentation', '1.0', config);
20
21        const myServiceMeterProvider = this.generateMeterForResource(resource);
22        metrics.setGlobalMeterProvider(myServiceMeterProvider);
23
24        this.cwvMeter = metrics.getMeter('core-web-vitals', '1.0.0');
25
26        // Initialising CWV metric instruments
27        this.lcp = this.cwvMeter.createObservableGauge('lcp', { unit: 'ms', description: 'Largest Contentful Paint' });
28        this.cls = this.cwvMeter.createObservableGauge('cls', { description: 'Cumulative Layout Shift' });
29        this.inp = this.cwvMeter.createObservableGauge('inp', { unit: 'ms', description: 'Interaction to Next Paint' });
30        this.ttfb = this.cwvMeter.createObservableGauge('ttfb', { unit: 'ms', description: 'Time to First Byte' });
31        this.fcp = this.cwvMeter.createObservableGauge('fcp', { unit: 'ms', description: 'First Contentful Paint' });
32    }
33
34    protected init(): InstrumentationModuleDefinition | InstrumentationModuleDefinition[] | void {}
35
36    // Creating Meter Provider factory to send metrics via OTLP
37    private generateMeterForResource(resource: Resource) {
38        const metricReader = new PeriodicExportingMetricReader({
39            exporter: new OTLPMetricExporter({
40                url: METRICS_URL //nginx proxy
41            }),
42            // Default is 60000ms (60 seconds).
43            // Set to 10 seconds for demo purposes only.
44            exportIntervalMillis: 10000
45        });
46
47        return new MeterProvider({
48            resource: resource,
49            readers: [metricReader]
50        });
```

```
53      enable() {
54          // Capture Largest Contentful Paint
55          onLCP(
56              (metric) => {
57                  this.lcp.addCallback((result) => {
58                      this.sendMetric(metric, result);
59                  });
60              },
61              { reportAllChanges: true }
62          );
63
64          // Capture Cumulative Layout Shift
65          onCLS(
66              (metric) => {
67                  this.cls.addCallback((result) => {
68                      this.sendMetric(metric, result);
69                  });
70              },
71              { reportAllChanges: true }
72          );
73
74          // Capture Interaction to Next Paint
75          onINP(
76              (metric) => {
77                  this.inp.addCallback((result) => {
78                      this.sendMetric(metric, result);
79                  });
80              },
81              { reportAllChanges: true }
82          );
83
84          // Time to First Byte
85          onTTFB(
```

```
90              },
91              { reportAllChanges: true }
92          );
93
94          // First Contentful Paint
95          onFCP(
96              (metric) => {
97                  this.fcp.addCallback((result) => {
98                      this.sendMetric(metric, result);
99                  });
100             },
101             { reportAllChanges: true }
102         );
103     }
104
105     private sendMetric(metric: CWVMetric, result: ObservableResult<Attributes>): void {
106         const now = hrTime();
107
108         const attributes = {
109             startTime: now,
110             'web_vital.name': metric.name,
111             'web_vital.id': metric.id,
112             'web_vital.navigationType': metric.navigationType,
113             'web_vital.delta': metric.delta,
114             'web_vital.value': metric.value,
115             'web_vital.rating': metric.rating,
116             // metric specific attributes
117             'web_vital.entries': JSON.stringify(metric.entries)
118         };
119
120         result.observe(metric.value, attributes);
121     }
122 }
```

OTel Core Web Vitals Summary

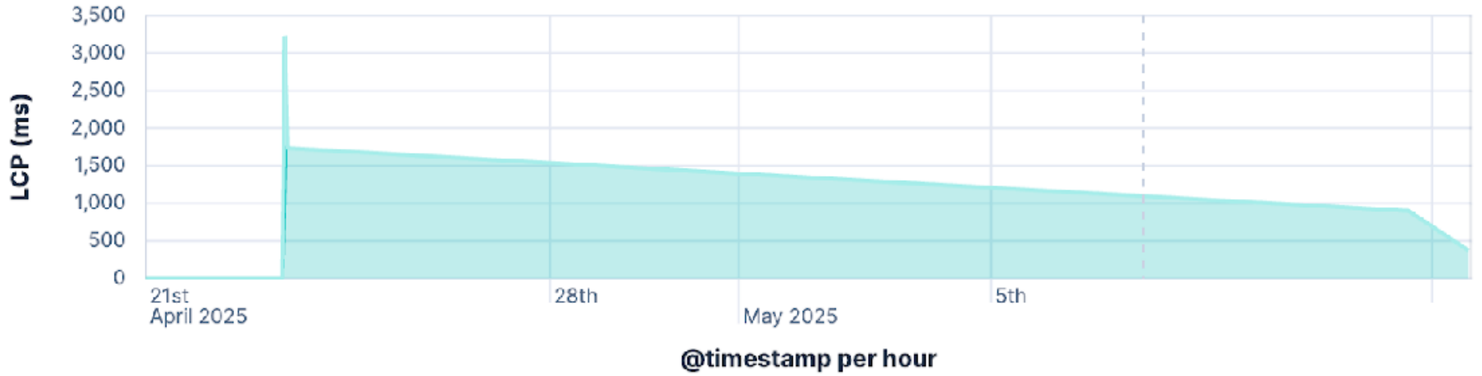
This dashboard shows the results of Core Web Vitals captured from the OTel record store application using the web-vitals JS library and OpenTelemetry JavaScript SDK.

Full instrumentation code available here.

Largest Contentful Paint (LCP)

good

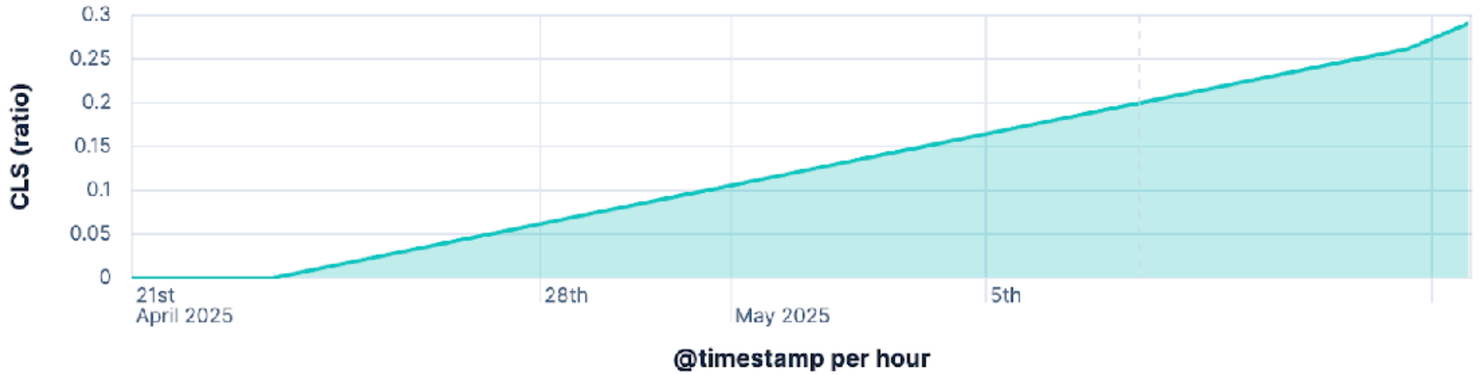
372



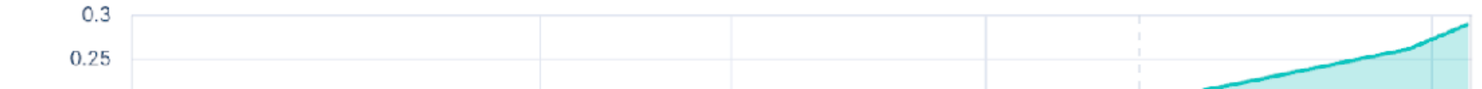
Cumulative Layout Shift (CLS)

poor

0.291



Interaction to Next Paint (INP)



Exit full screen

elastic

Source: Ruben. on Unsplash

Logs

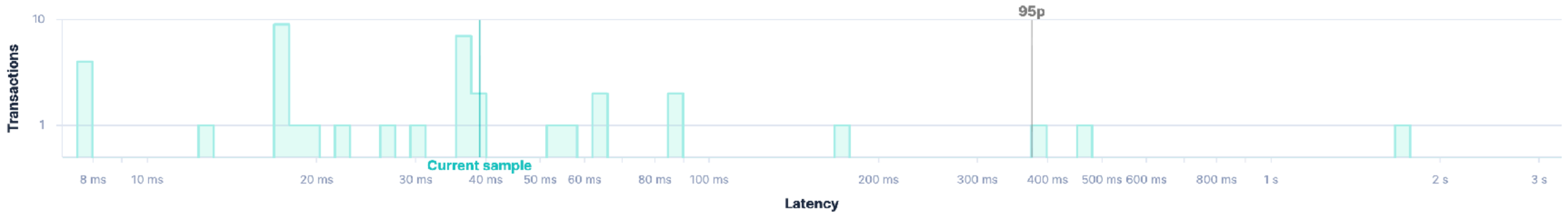
Metrics

Traces

Latency distribution

38 total transactions

Click and drag to select a range



All transactions

Trace sample

1 of 38

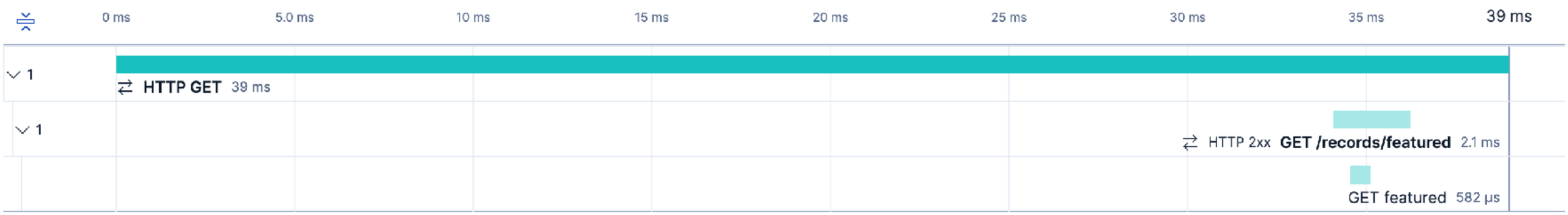
Investigate | View full trace

17 seconds ago | 39 ms (100% of trace)

Timeline | Metadata | Logs

Show critical path

Services: records-ui-web, record-store-server-java



```
28 const logger = logs.getLogger('default', '1.0.0');
29 logger.emit({
30     severityNumber: SeverityNumber.INFO,
31     severityText: 'INFO',
32     body: 'Logger initialized'
33 });
34
35 // Configure the OTLP exporter to talk to the collector via nginx
36 const exporter = new OTLPTraceExporter({
37     url: TRACE_URL // nginx proxy
38 });
39
40 // Instantiate the trace provider and inject the resource
41 const provider = new WebTracerProvider({
42     resource: resource,
43     spanProcessors: [
44         // Send each completed span through the exporter
45         new BatchSpanProcessor(exporter),
46         new BatchSpanProcessor(new ConsoleSpanExporter())
47     ]
48 });
49
50 // Register the provider with propagation and set up the async context manager for spans
51 provider.register({
52     contextManager: new ZoneContextManager(),
53     propagator: new CompositePropagator({
54         propagators: [new W3CBaggagePropagator(), new W3CTraceContextPropagator()]
55     })
56 });
57
```

```
39
40 // Instantiate the trace provider and inject the resource
41 const provider = new WebTracerProvider({
42     resource: resource,
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52     contextManager: new ZoneContextManager(),
53     propagator: new CompositePropagator({
54         propagators: [new W3CBaggagePropagator(), new W3CTraceContextPropagator()]
55     })
56 });
57
58 export class ClientTelemetry {
59     private static instance: ClientTelemetry;
60     private initialized = false;
61
62     private constructor() {}
63
64     // Obtain singleton instance of provider
65     public static getInstance(): ClientTelemetry {
66         if (!ClientTelemetry.instance) {
67             ClientTelemetry.instance = new ClientTelemetry();
68             ClientTelemetry.instance.start();
```
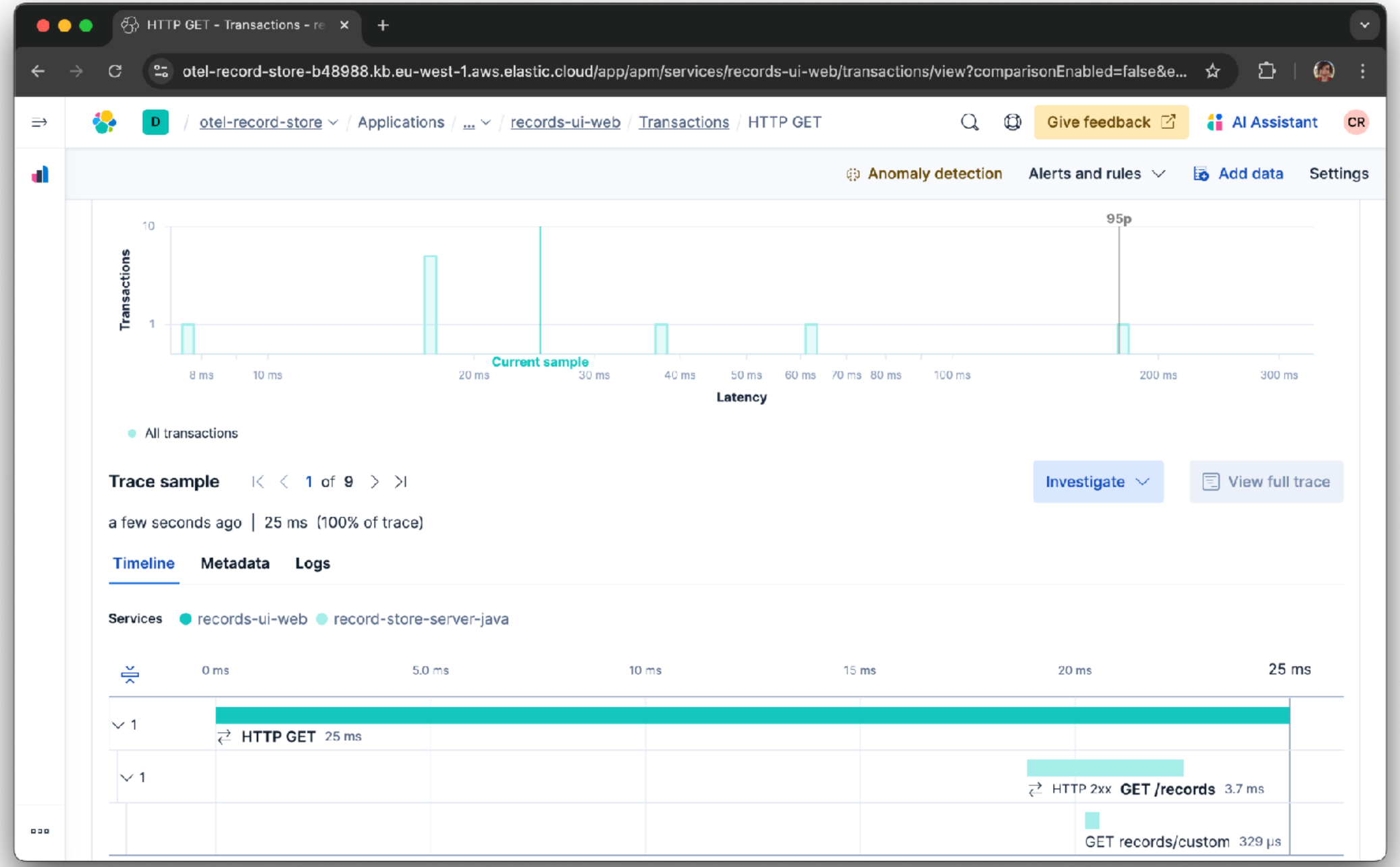

Source: engin akyurt on Unsplash


HTTP GET - Transactions - re
otel-record-store-b48988.kb.eu-west-1.aws.elastic.cloud/app/apm/services/records-ui-web/transactions/view?comparisonEnabled=false&e...
D
otel-record-store / Applications / ... / records-ui-web / Transactions / HTTP GET
Give feedback
AI Assistant
CR
Anomaly detection
Alerts and rules
Add data
Settings
95p
Transactions
10
1
Current sample
8 ms
10 ms
20 ms
30 ms
40 ms
50 ms
60 ms
70 ms
80 ms
100 ms
200 ms
300 ms
Latency
All transactions
Trace sample
1 of 9
Investigate
View full trace
a few seconds ago | 25 ms (100% of trace)
Timeline
Metadata
Logs
Services
records-ui-web
record-store-server-java
0 ms
5.0 ms
10 ms
15 ms
20 ms
25 ms
1
HTTP GET 25 ms
1
HTTP 2xx GET /records 3.7 ms
GET records/custom 329 µs

frontend.tracer.ts

```
/* OpenTelemetry Frontend packages */

// See https://github.com/carlyrichmond/otel-record-store for import list

// Define resource metadata for the service, used by exporters (Elastic requires service.name)
const SERVICE_NAME = 'records-ui-web';

const detectedResources = detectResources({ detectors: [browserDetector] });
let resource = resourceFromAttributes({
    [ATTR_SERVICE_NAME]: SERVICE_NAME,
    'service.version': 1,
    'deployment.environment': 'dev'
});
resource = resource.merge(detectedResources);

// Configure logging to send to the collector via nginx
const logExporter = new OTLPLogExporter({
    url: LOGS_URL // nginx proxy
});

const loggerProvider = new LoggerProvider({
    resource: resource,
    processors: [new BatchLogRecordProcessor(logExporter)]
});

logs.setGlobalLoggerProvider(loggerProvider);

```

```
72
73      // Initializer
74      public start() {
75          if (!this.initialized) {
76              // Enable automatic span generation for document load and user click interactions
77              registerInstrumentations({
78                  instrumentations: [
79                      // Automatically tracks when the document loads
80                      new DocumentLoadInstrumentation({
81                          ignoreNetworkEvents: false,
82                          ignorePerformancePaintEvents: false
83                      }),
84                      getWebAutoInstrumentations({
85                          '@opentelemetry/instrumentation-fetch': {
86                              propagateTraceHeaderCorsUrls: /.*/,
87                              clearTimingResources: true
88                          },
89                          '@opentelemetry/instrumentation-xml-http-request': {
90                              propagateTraceHeaderCorsUrls: /.*/
91                          }
92                      }),
93                      // User events
94                      new UserInteractionInstrumentation({
95                          eventNames: ['click', 'input'] // instrument click and input events only
96                      }),
97                      // Custom Web Vitals instrumentation
98                      new WebVitalsInstrumentation({}, resource)
99                  ]
100             });
101
```

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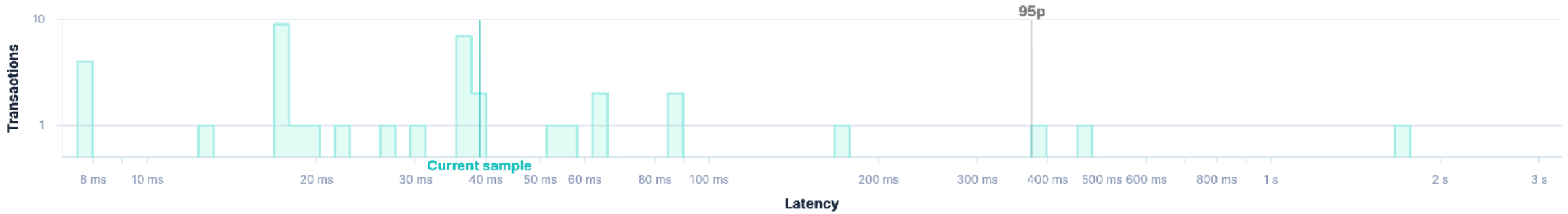
100             });
101
```

Latency distribution

38 total transactions

Click and drag to select a range



All transactions

Trace sample

1 of 38

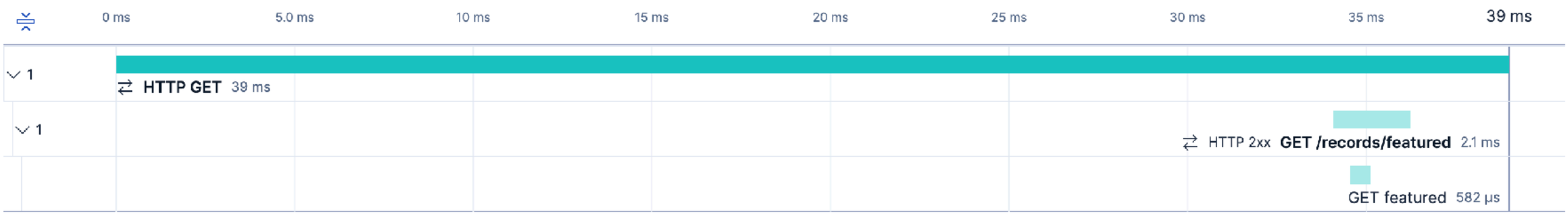
Investigate | View full trace

17 seconds ago | 39 ms (100% of trace)

Timeline | Metadata | Logs

Show critical path

Services: records-ui-web, record-store-server-java



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78                  instrumentations: [
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82                          ignorePerformancePaintEvents: false
83                      }),
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85                          '@opentelemetry/instrumentation-fetch': {
86                              propagateTraceHeaderCorsUrls: /.*/,
87                              clearTimingResources: true
88                          },
89                          '@opentelemetry/instrumentation-xml-http-request': {
90                              propagateTraceHeaderCorsUrls: /.*/
91                          }
92                      }),
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94                      new UserInteractionInstrumentation({
95                          eventNames: ['click', 'input'] // instrument click and input events only
96                      }),
97                      // Custom Web Vitals instrumentation
98                      new WebVitalsInstrumentation({}, resource)
99                  ]
100             });
101
```

Latency

All transactions

Trace sample 1 of 9

Investigate | View full trace

a month ago | 230 ms (100% of trace)

Timeline | Metadata | Logs

Show critical path

Type: records-ui-web, http

All transactions

Trace sample 1 of 9

a month ago | 230 ms (100% of trace)

Timeline Metadata Logs

Show critical path

Type records-ui-web http

0 s 50 s

7 Success documentLoad 230 ms

Span details

View span in Discover

Name	Dependency	Service	Transaction
resourceFetch	localhost:4173	records-ui-web	documentLoad

a month ago | 11 ms (0.0% of transaction) | external http

Metadata

Filter...

How to add labels and other data

@timestamp

@timestamp	2025-07-25T12:09:58.941Z

agent

agent.name	otlp
agent.version	unknown

attributes

attributes.event.outcome	success
attributes.event.success_count	1
attributes.http.response_content_length	13262
attributes.http.url	http://localhost:4173/src/app.css?t=1753445398940
attributes.processor.event	span
attributes.service.target.name	localhost:4173
attributes.service.target.name.text	localhost:4173
attributes.service.target.type	http

```
72
73      // Initializer
74      public start() {
75          if (!this.initialized) {
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77              registerInstrumentations({
78                  instrumentations: [
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99                  ]
100             });
101
```
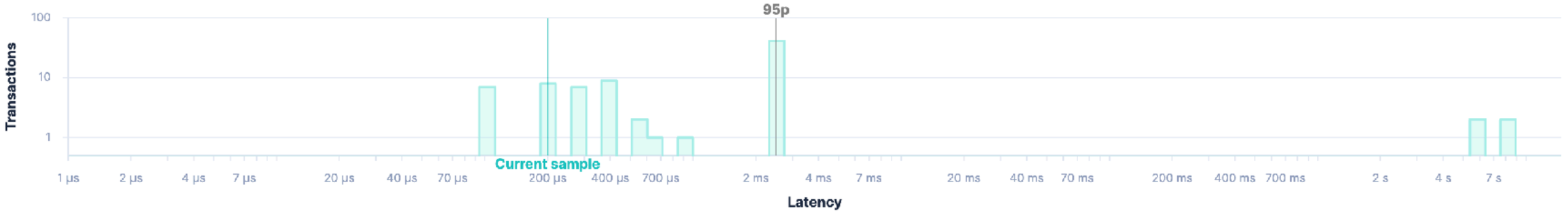

Trace samples | Latency correlations | Failed transaction correlations

Latency distribution

85 total transactions

Click and drag to select a range



All transactions

Trace sample

1 of 85

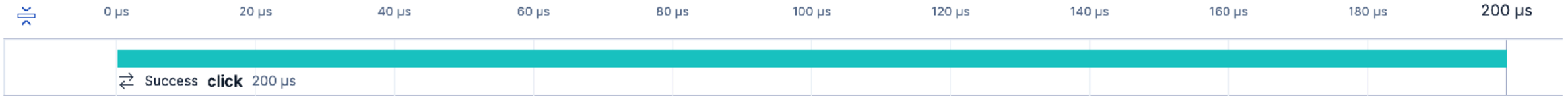
Investigate | View full trace

19 minutes ago | 200 µs (100% of trace)

Timeline | Metadata | Logs

Show critical path

Type records-ui-web

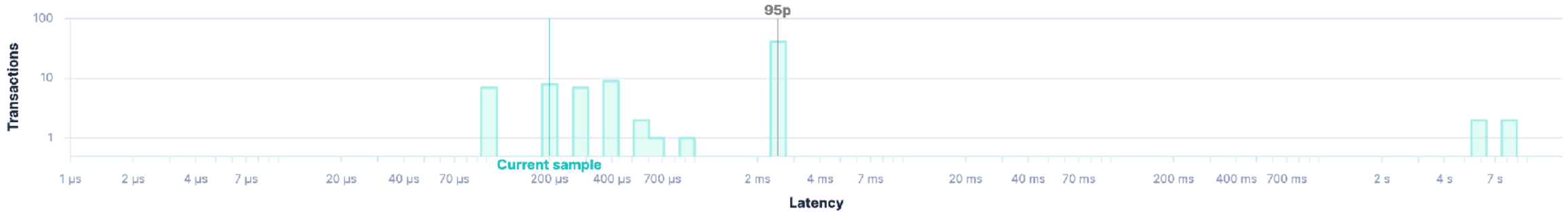


Trace samples | Latency correlations | Failed transaction correlations

Latency distribution

85 total transactions

Click and drag to select a range



All transactions

Trace sample

1 of 85

Investigate | View full trace

20 minutes ago | 200 µs (100% of trace)

Timeline | Metadata | Logs

xpath

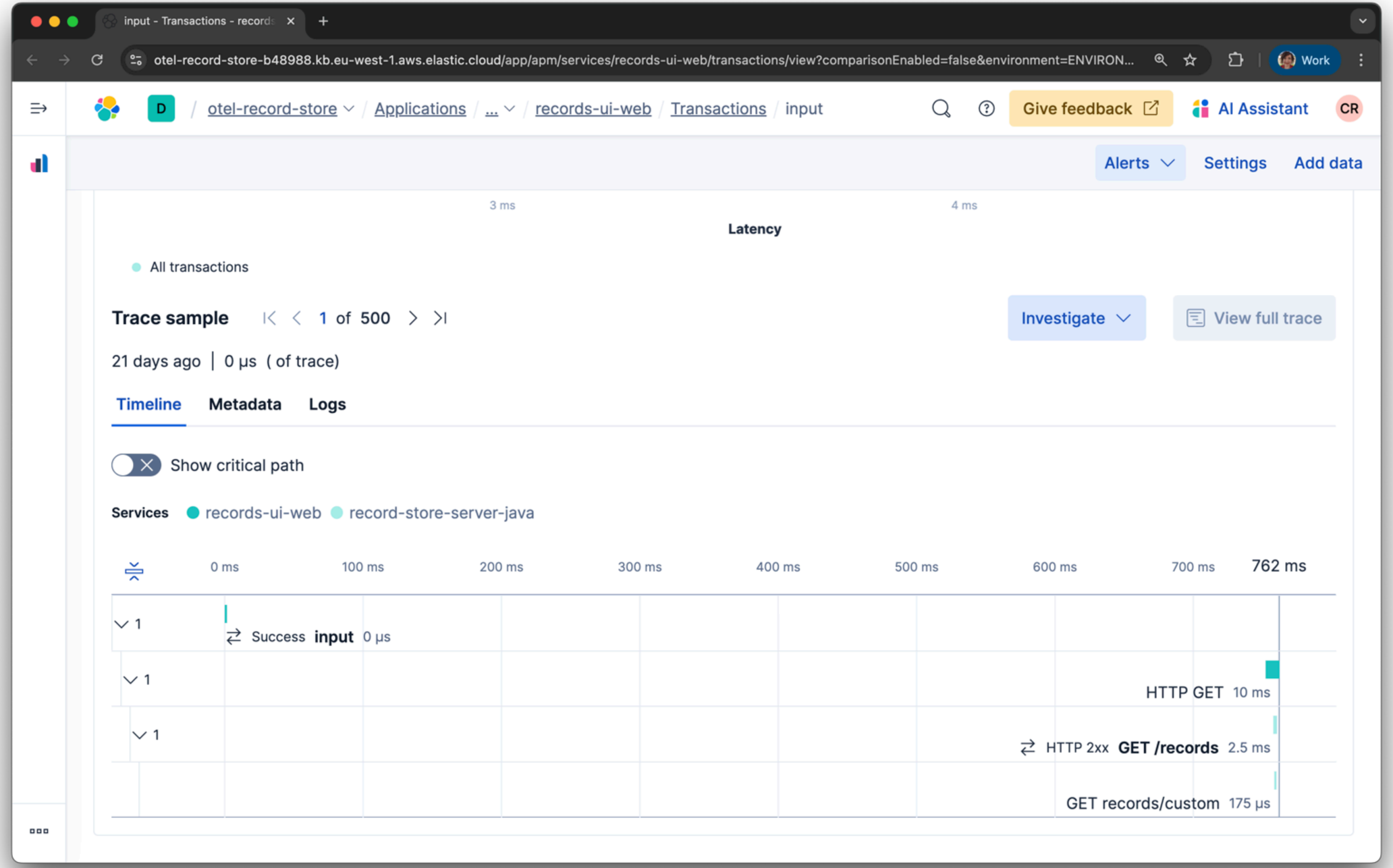
How to add labels and other data

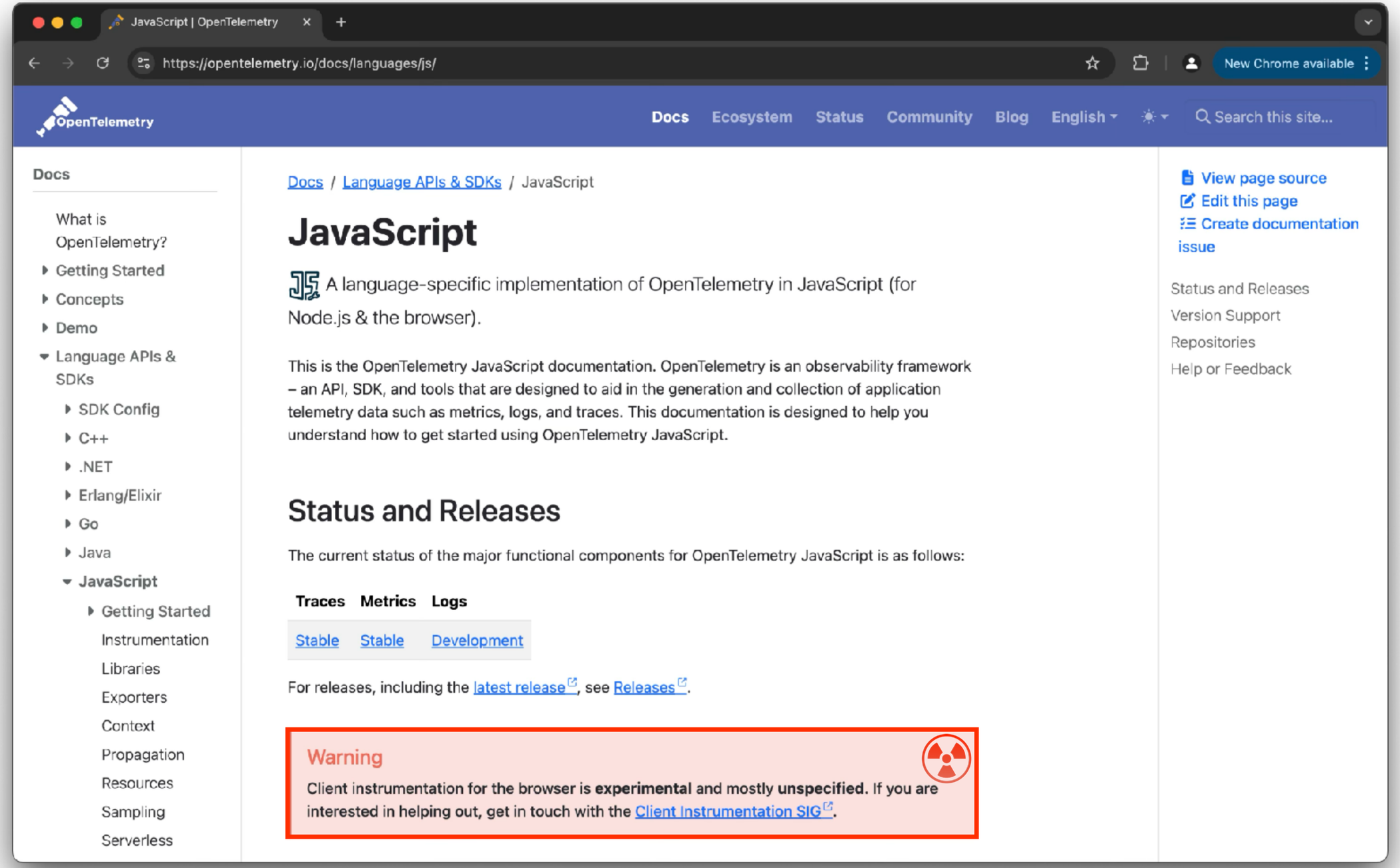
attributes

attributes.target_xpath	//html/body/div/section/div/button

target_xpath

target_xpath	//html/body/div/section/div/button


input - Transactions - records
otel-record-store-b48988.kb.eu-west-1.aws.elastic.cloud/app/apm/services/records-ui-web/transactions/view?comparisonEnabled=false&environment=ENVIRON...
Work
D
otel-record-store
Applications
...
records-ui-web
Transactions
input
Give feedback
AI Assistant
CR
Alerts
Settings
Add data
3 ms
4 ms
Latency
All transactions
Trace sample
1 of 500
Investigate
View full trace
21 days ago | 0 µs (of trace)
Timeline
Metadata
Logs
Show critical path
Services
records-ui-web
record-store-server-java
0 ms
100 ms
200 ms
300 ms
400 ms
500 ms
600 ms
700 ms
762 ms
1
Success input 0 µs
1
HTTP GET 10 ms
1
HTTP 2xx GET /records 2.5 ms
GET records/custom 175 µs


JavaScript | OpenTelemetry
https://opentelemetry.io/docs/languages/js/
New Chrome available
OpenTelemetry
Docs
Ecosystem
Status
Community
Blog
English
Search this site...
Docs
What is OpenTelemetry?
Getting Started
Concepts
Demo
Language APIs & SDKs
SDK Config
C++
.NET
Erlang/Elixir
Go
Java
JavaScript
Getting Started
Instrumentation
Libraries
Exporters
Context
Propagation
Resources
Sampling
Serverless
Docs / Language APIs & SDKs / JavaScript
JavaScript
A language-specific implementation of OpenTelemetry in JavaScript (for Node.js & the browser).
This is the OpenTelemetry JavaScript documentation. OpenTelemetry is an observability framework – an API, SDK, and tools that are designed to aid in the generation and collection of application telemetry data such as metrics, logs, and traces. This documentation is designed to help you understand how to get started using OpenTelemetry JavaScript.
Status and Releases
The current status of the major functional components for OpenTelemetry JavaScript is as follows:
Traces
Metrics
Logs
Stable
Stable
Development
For releases, including the latest release, see Releases.
Warning
Client instrumentation for the browser is experimental and mostly unspecified. If you are interested in helping out, get in touch with the Client Instrumentation SIG.
View page source
Edit this page
Create documentation issue
Status and Releases
Version Support
Repositories
Help or Feedback

Source: Ruben. on Unsplash

Logs

Metrics

Traces

Thank you!